

LECTURE 1 (SEPTEMBER 2ND)

Definition. (Contour integral) Let C be a contour parametrised by $z = z(t)$ where $a \leq t \leq b$. Also, let f be continuous on C . Then, the contour integral of f is defined as

$$\int_C f(z) dz = \int_C u dx - v dy + i \int_C u dy + v dx$$

and also, through parametrisation, we have

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

Definition. (Derivative) The derivative of a complex function is defined as

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

if it exists. If f is C^1 (the function is continuous up its first derivatives) and $u_x = v_y$ & $v_x = -u_y$, then $f'(z)$ exists.

Theorem. (Green's theorem) Let P and Q be a C^1 function on $\bar{D}$ where D is a simply connected bounded region whose boundary is C . Then,

$$\int_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

where C is oriented in the positive sense. Here,

$$\int_C P dx = - \iint_D \frac{\partial Q}{\partial y} dy dx$$

and

$$\int_C Q dy = \iint_D \frac{\partial Q}{\partial x} dx dy$$

Proof. Notice that for a parametrised simple disc where the above is $y = g_2(x)$ and $y = g_1(x)$ on the bottom,

$$\int_C P dx = \int_a^b P(x, g_1(x)) dx - \int_a^b P(x, g_2(x)) dx$$

and

$$\iint_D \frac{\partial P}{\partial y} dy dx = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y} dy dx = \int_a^b \left[P(x, g_2(x)) - P(x, g_1(x)) \right] dx$$

Similarly,

$$\int_D Q \, dy = \int_c^d Q(y, h_2(x)) \, dy - \int_c^d Q(h_1(y), y) \, dy$$

and

$$\iint \frac{\partial Q}{\partial x} \, dx \, dy = \int_c^d \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x} \, dx \, dy = \left[Q(h_2(y), y) - Q(h_1(y), y) \right] dy$$

□

Theorem. (Cauchy's theorem) If f is analytic (differentiable) inside and on a bounded simply connected domain D whose boundary is C and $f \in C^1(\bar{D})$ then

$$\int_C f(z) \, dz = 0$$

Proof. Let $f = u + iv$. Then $u_x = v_y$ and $v_x = -u_y$ on D . Thus

$$\int_C f(z) \, dz = \int_C u \, dx - v \, dy + i \left[\int_C u \, dy + v \, dx \right]$$

where

$$\int_C u \, dx - v \, dy = \iint_D \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

and

$$\int_C u \, dy + v \, dx = \iint_D \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \, dx \, dy$$

notice how the terms in the large brackets vanish, and that the integral itself equals zero. □

Theorem. (Generalised Cauchy's Theorem) If $f \in C^1(\bar{D})$ is analytic in a bounded multiply connected domain D with finitely many holes inside, then

$$\int_{\partial D} f(z) \, dz = 0$$

where ∂D is a positively oriented boundary of D .

Theorem. (Cauchy Integral Formula) If f is analytic inside and on a POSCC C then for every z_0 inside C we have

$$f(z_0) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} \, dz$$

LECTURE 2 (SEPTEMBER 4TH)

Recall. (Cauchy's formula) For a positively oriented simple closed curve (POSCC) C and f that is analytic inside and on C , we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{\xi - z} d\xi$$

for all z inside C . If $\bar{D}(z_0, r) = \{z \mid |z - z_0| \leq r\} \subset D$,

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0 + z_0 - z} = \frac{1}{(\xi - z_0) \left(1 - \frac{z - z_0}{\xi - z_0}\right)}$$

for $z \in D(z_0, r)$. However,

$$\left| \frac{z - z_0}{\xi - z_0} \right| < \frac{|z - z_0|}{r} < 1$$

so that

$$\frac{1}{\xi - z} = \frac{1}{\xi - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{\xi - z_0} \right)^n = \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}}$$

Thus

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \int_C f(\xi) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\xi - z_0)^{n+1}} d\xi \\ &= \sum_{n=0}^{\infty} a_n (z - z_0)^n \end{aligned}$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(\xi)}{(\xi - z_0)^{n+1}} d\xi = \frac{1}{n!} f^{(n)}(z_0)$$

Notice that the convergence is uniform for $\xi \in C$.

Theorem. If f_n is continuous in C such that $f_n \rightarrow f$ uniformly on C

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz$$

Theorem. If $f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n$ for $z \in D(z_0, R)$ then for $0 < r < R$ we have

$$\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt = \sum_{n=0}^{\infty} |c_n|^2 r^{2n}$$

Proof. Take

$$f(z_0 + re^{it}) = \sum_{n=0}^{\infty} c_n r^n e^{int}$$

so that

$$\begin{aligned}
\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})|^2 dt &= \frac{1}{2\pi} \int_0^{2\pi} \sum_{n=0}^{\infty} c_n r^n e^{int} \sum_{m=0}^{\infty} \bar{c}_m r^m e^{-imt} dt \\
&= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{1}{2\pi} \int_0^{2\pi} c_n \bar{c}_m r^{n+m} e^{i(n-m)t} dt \\
&= \sum_{n=0}^{\infty} |c_n|^2 r^{2n}
\end{aligned}$$

Here, we note that

$$\int_0^{2\pi} e^{i(n-m)t} dt = \begin{cases} 2\pi & n = m \\ 0 & n \neq m \end{cases}$$

□

Corollary. (Liouville theorem) All bounded entire functions are constant. Using this theorem, we can prove the fundamental theorem of algebra.

Proof. If f is entire such that $|f(z)| \leq M$ for all $z \in \mathbf{C}$ then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq M^2$$

for all $r > 0$ which means that $c_1 = c_2 = \dots = 0$.

□

Theorem. (Identity theorem) Let D be a domain (open and connected) and $\{z_n\}$ be a sequence of distinct points in D such that $z_n \rightarrow z_0 \in D$. If f is analytic in D with $f(z_n) = 0$ for all $n \in \mathbf{N}$, then $f(z) = 0$ for all $z \in D$.

Theorem. (Maximum modulus theorem) Let f be analytic in a domain D . Then, $|f|$ can not have a local maximum in D unless it is constant.

Proof. If

$$|f(z_0 + re^{it})| \leq |f(z_0)|$$

for all $t \in [0, 2\pi]$, then

$$\sum_{n=0}^{\infty} |c_n|^2 r^{2n} \leq |f(0)|^2 = |c_0|^2$$

which means that $c_1 = \dots = c_n = 0$. Then we can apply the identity theorem.

□

Theorem. (Morera's theorem) Morera's theorem can be thought as Cauchy theorem's inverse. If f is continuous in a domain D such that

$$\int_C f(z) dz = 0$$

for all closed contours C in D , then f is analytic in D (note that this also means that the integral is path independent).

Proof. The key point is that for a given $a \in D$, $F(z) = \int_a^z f(\xi) d\xi$ is well defined. Take any $z_0 \in D$. We'll show that $F'(z_0) = f(z_0)$, which means that $f = F'$ is analytic. Take any $\varepsilon > 0$. We choose $\delta > 0$ so that

(i) $D(z_0, \delta) \subset D$

(ii) If $|z - z_0| < \delta$ then $|f(z) - f(z_0)| < \varepsilon$

We need to prove that if $0 < |z - z_0| < \delta$ then

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \varepsilon$$

Take the line segment path $[z_0, z]$ from z_0 to z . Then

$$F(z) - F(z_0) = \int_{[z_0, z]} f(\xi) d\xi$$

and

$$\frac{F(z) - F(z_0)}{z - z_0} - f(z_0) = \frac{1}{z - z_0} \int_{[z_0, z]} [f(\xi) - f(z_0)] d\xi$$

Notice that now, due to (ii), the term inside the square brackets can be reduced to less than ε since $|\xi - z_0| < \delta$. Therefore

$$\left| \frac{F(z) - F(z_0)}{z - z_0} - f(z_0) \right| < \frac{1}{|z - z_0|} |z - z_0| \varepsilon = \varepsilon$$

□

Theorem. Let $\{f_n\}$ be a sequence of analytic functions in a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D . Then f is analytic on D .

Example. On $\mathbf{R}$, consider

$$f_n(x) = \sum_{k=1}^n \frac{\cos 3^k x}{2^k} \rightarrow f(x) = \sum_{n=1}^{\infty} \frac{\cos 3^n x}{2^n}$$

where the convergence is uniform and the right-hand side is nowhere differentiable.

Proof. First, f is continuous on D since convergence is uniform. Let C be any closed contour in D . Then

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \int_C f_n(z) dz = 0$$

By Morera's theorem, the proof is complete. $\square$

LECTURE 3 (SEPTEMBER 9TH)

Theorem. Let f be analytic on a simply connected domain D such that $f(z) \neq 0$ for any $z \in D$. Then,

- (i) There is g analytic on D such that $e^g = f$ on D
- (ii) There is h analytic on D such that $h^2 = f$ on D (this implies the Riemann mapping theorem)

Proof. (1) By Cauchy's theorem, the $\int_C f(z) dz = 0$ for every closed contour C in D . Thus for a given $z_0 \in D$, the integral

$$\int_{z_0}^z f(\xi) d\xi$$

is independent of path. Since f has no zero in D , f'/f is analytic on D . We define g on D by

$$g(z) = \int_{z_0}^z \frac{f'(\xi)}{f(\xi)} d\xi + \text{Log } f(z_0)$$

Then we'll show that $f(z) = e^{g(z)}$ on D . Observe that

$$\begin{aligned} (fe^{-g})' &= f'e^{-g} + f(e^{-g})' \\ &= f'e^{-g} + fe^{-g}\left(-\frac{f'}{f}\right) \\ &= f'e^{-g} - f'e^{-g} = 0 \end{aligned}$$

on D and $fe^{-g} = c$ a constant. If we define $h(z) = f(z)e^{-g(z)}$ then

$$\begin{aligned} h(z_0) &= f(z_0)e^{-g(z_0)} \\ &= f(z_0)e^{-\text{Log } f(z_0)} \\ &= f(z_0)[f(z_0)]^{-1} = 1 \end{aligned}$$

(2) By (1), there exists an analytic g such that $f = e^g$ on D . If we define $h = e^{g/2}$, then h is analytic on D with $h^2 = f$. $\square$

Theorem. (Riemann mapping theorem) If D is a simply connected domain in $\mathbf{C}$ ($D \neq \mathbf{C}$), then there is a one-to-one analytic function f from D onto $U = \{z \in \mathbf{C} \mid |z| < 1\}$ (the unit disk).

Proof. The proof consists of three steps. Let Σ be a set of all 1-1 analytic functions from D to U .

$$\Sigma = \{f : D \rightarrow U \mid f \text{ is 1-1 analytic}\}$$

We first prove that

- (i) $\Sigma \neq \emptyset$
- (ii) That if $\psi \in \Sigma$ satisfies $\psi(D) \neq U$, then for every $z_0 \in D$ there is $\psi_0 \in \Sigma$ such that $|\psi'_0(z_0)| > |\psi'(z_0)|$,
- (iii) For a given $z_0 \in D$ if $\eta_0 = \sup\{|\psi'(z_0)| \mid \psi \in \Sigma\}$ then there is $f \in \Sigma$ such that $|f'(z_0)| = \eta_0$.

□

Theorem. (Exhaustion of D by compact subsets) Let $D \subset \mathbf{C}$ be open. Then there is a sequence of compact subsets K_n of D such that

- (i)

$$K_1 \subset K_2 \subset \dots \subset K_n \subset D$$

- (ii)

$$\bigcup_{n=1}^{\infty} K_n = D$$

- (iii) If $K \subset D$ is compact then $K \subset K_n$ for some $n \in \mathbf{N}$

- (iv) There is a sequence $\{\delta_n\}$ with $\delta_n > 0$ such that

$$\bar{D}(z, 2\delta_n) \subset K_{n+1}$$

for each $z \in K_n$

Proof. Define

$$K_n = \left\{z \in \mathbf{C} \mid |z| \leq n\right\} \cap \left\{z \in \mathbf{C} \mid d(z, D^c) \geq \frac{1}{n}\right\}$$

$$G_n = \left\{|z| < n\right\} \cap \left\{d(z, D^c) > \frac{1}{n}\right\}$$

and $G_n = D$ by the Archimedean property.

□

LECTURE 4 (SEPTEMBER 16TH)

Theorem. Let D be a domain in $\mathbf{C}$. There is a sequence $\{K_n\}$ of compact subsets of D and $\{\delta_n\}$ ($\delta_n > 0$) such that

(i)

$$\bigcup_n^\infty K_n = D$$

(ii) If $K \subset D$ is compact then $K \subset K_n$ for some n

(iii) $\bar{D}(z, 2\delta_n) \subset K_{n+1}$ for every $z \in K_n$, $n = 1, 2, \dots$

Note that $z \notin F$ and F is closed. This implies that $d(z, F) = \min\{|z - w| \mid w \in F\} > 0$.

Proof. Put

$$K_n = \left\{ z \leq n \right\} \cap \left\{ d(z, D^C) \geq \frac{1}{n} \right\}$$

□

Example. Consider

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \rightarrow 0$$

which converges uniformly on $\mathbf{R}$. However,

$$f'_n(0) = \sqrt{n} \rightarrow \infty$$

as $n \rightarrow \infty$.

Theorem. If $\{f_n\}$ is a sequence of analytic functions on a domain D such that $f_n \rightarrow f$ uniformly on every compact subset of D , then

(i) f is analytic on D (by Morera's theorem)

(ii) $f'_n \rightarrow f'$ uniformly on every compact subset of D

Proof. Take any compact subset K of D . Then $K \subset K_n$ for some n , and $\bar{D}(z, \delta_n) \subset K_{n+1}$ for all $z \in K$. So, there exists $r > 0$ and K' , a compact subset of D , such that $\bar{D}(z, r) \subset K'$ for all $z \in K$. Here, applying Cauchy's integral formula,

$$f(z) = \frac{1}{2\pi i} \int_{|\xi-z|=r} \frac{f(\xi)}{(\xi-z)^2} d\xi$$

For every $z \in K$, we have

$$f'(z) = \frac{1}{2\pi i} \int_{|\xi-z|=r} \frac{f(\xi)}{(\xi-z)^2} d\xi$$

which implies that

$$|f'(z)| \leq \frac{1}{2\pi} \frac{\|f\|_{K'}}{r^2} (2\pi r) = \frac{\|f\|_{K'}}{r}$$

with $\|f\|_K = \max_{z \in K} |f(z)|$. Therefore, for every $z \in K$,

$$|f'_n(z) - f'(z)| \leq \frac{\|f_n - f\|_{K'}}{r}$$

which means $f'_n \rightarrow f'$ uniformly on K . □

Example. (Linear mapping) Consider the linear mapping

$$w = az + b$$

This mapping we see that, by expressing $a = re^{i\theta}$, the map stretches, rotates, and translates the complex plane.

Definition. (Linear functional transformation) For $ad - bc \neq 0$,

$$w = T(z) = \frac{az + b}{cz + d}$$

This can be written as

$$w = T(z) = \frac{a}{c} + \frac{bc - ad}{c} \cdot \frac{1}{cz + d}$$

Additionally, the map is 1-1 from the fact that the following inverse map exists:

$$z = \frac{dw - b}{-cw + a}$$

Thus telling us that the map can be seen as a combination of translation, dilation ($|a| > 1$) or contraction ($|a| \leq 1$), rotation, and inversion.

$$z \rightarrow \frac{1}{z}$$

This is the tricky one!

Remark. (Inversion map $w = 1/z$) The equation (locus) of all lines and all circles in $\mathbf{R}^2$ is

$$a(x^2 + y^2) + bx + cy + d = 0$$

where $b^2 + c^2 > 4ad$ and $a, b, c, d \in \mathbf{R}$. In the complex plane, for $x^2 + y^2 = |z|^2$, $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/2i$ so that

$$a|z|^2 + \frac{1}{2}(b - ic)z + \frac{1}{2}(b + ic)\bar{z} + d = 0$$

and again $b^2 + c^2 > 4ad$. This implies that

$$a|z|^2 + \beta z + \bar{\beta}\bar{z} + d = 0$$

where $a, d \in \mathbf{R}$ and $\beta \in \mathbf{C}$ with $|\beta|^2 > ad$. Taking the inversion map,

$$a \frac{1}{|w|^2} + \beta \frac{1}{w} + \bar{\beta} \frac{1}{\bar{w}} + d|w|^2 = 0$$

and

$$a + \beta\bar{w} + \bar{\beta}w + d = 0$$

by multiplying $|w|^2 = w\bar{w}$ ($\beta\bar{\beta} > ad$). We therefore see that the inversion map takes the set of all lines and circles to again the set of all lines and circles.

Definition. (S^2) S^2 is defined as the boundary of a unit ball in $\mathbf{R}^3$. We see this as a one-point compactification (Alexandroff compactification) of $\mathbf{C}$, and is seen as $\mathbf{C} \cup \{\infty\}$ where $\{\infty\}$ is called the north pole.

Lemma. The linear functional transformation which takes $\{a, b, c\}$ to $\{0, 1, \infty\}$ is uniquely determined by

$$T(z) = \frac{(z-a)(b-c)}{(z-c)(b-a)}$$

LECTURE 5 (SEPTEMBER 18TH)

Definition. Notice that the linear fractional transformation defined for $ad - bc \neq 0$ as

$$T(z) = \frac{az + b}{cz + d}$$

we see that the derivative is

$$T'(z) = \frac{ad - bc}{(cz + d)^2} \neq 0$$

and that it is analytic and T' is never 0. Meanwhile,

$$T^{-1}(z) = \frac{dz - b}{-cz + a}$$

Conformal are by definition, angle preserving maps (both size and orientation).

Definition. Let $z = z(t)$ for $a \leq t \leq b$ represent the contour C . Also, let $w = f(z)$, that is, $w(t) = f(z(t))$ represent the contour Γ . If $z_0 = z(t_0)$ and $w_0 = f(z_0) = f(z(t_0))$, f is analytic at z_0 such that $f'(z_0) \neq 0$. Then, from

$$w'(t_0) = f'(z_0)z'(t_0)$$

we have

$$\arg w'(t_0) = \arg f'(z_0) + \arg z'(t_0)$$

Thus, we see that analytic functions with nonzero derivatives are conformal mappings.

Definition. Consider

$$T(z) = \frac{az + b}{cz + d}$$

(with $ad - bc \neq 0$) and

$$S(z) = \frac{ez + f}{gz + h}$$

(with $eh - gf \neq 0$) we ultimately have

$$(T \circ S)(z) = \frac{(ae + bg)z + (af + bh)}{(ce + dg)z + (cf + dh)}$$

also,

$$T^{-1}(z) = \frac{dz - b}{-cz + a}$$

Ultimately, we see that these mappings are equivalent to matrix multiplication.

Proposition. (The fixed points of $T(z) = z$) We see that

$$\frac{az + b}{cz + d} = z \quad \implies \quad cz^2 + (d - a)z - b = 0$$

and that there can't be more than two fixed points. If S and T are linear fractional transformations such that $S(z_k) = T(z_k)$ for $k = 1, 2, 3$, then $S = T$ since $S^{-1} \circ T = I$.

Theorem. If distinct points $\{z_1, z_2, z_3\}$ and distinct points $\{w_1, w_2, w_3\}$ are points of $S^2 = \mathbf{C} \cup \{\infty\}$ then there exists a unique linear fractional transformation T such that $T(z_k) = w_k$ for $k = 1, 2, 3$.

$$\{a, b, c\} \rightarrow \{0, 1, \infty\} \leftrightarrow \{\alpha, \beta, \gamma\}$$

Then

$$T(z) = \frac{(b - c)(z - a)}{(b - a)(z - c)} \quad \& \quad S(z) = \frac{(\beta - \gamma)(z - \alpha)}{(\beta - \alpha)(z - \gamma)}$$

Example. Find V such that

$$\{0, 1, 2\} \rightarrow \{-1, 0, 4\}$$

Proof.

$$S(z) = \frac{-4(z + 1)}{1(z - 4)} = \frac{-4z - 4}{z - 4}$$

(we say it maps to $\{0, 1, \infty\}$) which is equivalent to

$$\begin{pmatrix} -4 & -4 \\ 1 & -4 \end{pmatrix}$$

For T that sends $\{0, 1, 2\}$ to the same points, we find the matrix, and $V(z)$ is given as

$$V(z) = \frac{8z - 8}{-3z + 8}$$

□

Example. Let

$$T(z) = \frac{1+z}{1-z}$$

which corresponds to

$$\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

Let $U = \{|z| < 1\}$, and say that we want to find $T(U)$. Notice that

$$\{-1, 0, 1\} \mapsto \{0, 1, \infty\}$$

As $T([-1, 1])$ maps to the positive x -axis, we can see that $T(-1) = \infty$ and that $T(|z| = 1)$ is a line passing $T(-1) = 0$. Then, $T(i) = i$ and $T(|z| = 1)$ is the y -axis. Another way to do this is through conformality of $T(|z| = 1)$ and the positive real axis being $\pi/2$ at $z = 0$.

Definition. Let U be the unit disk and $T = \{|z| = 1\}$ be a unit circle. For $\alpha \in U$, we define

$$\phi_\alpha(z) = \frac{z - \alpha}{1 - \bar{\alpha}z} : \begin{pmatrix} 1 & -\alpha \\ -\bar{\alpha} & 1 \end{pmatrix}$$

Then ϕ_α is analytic on $\bar{U}$ such that $\phi_\alpha^{-1} = \phi_{-\alpha}$, $\phi_\alpha(0) = -\alpha$, $\phi_\alpha(\alpha) = 0$, $\phi_\alpha(U) = U$, and $\phi_\alpha(T) = T$.